

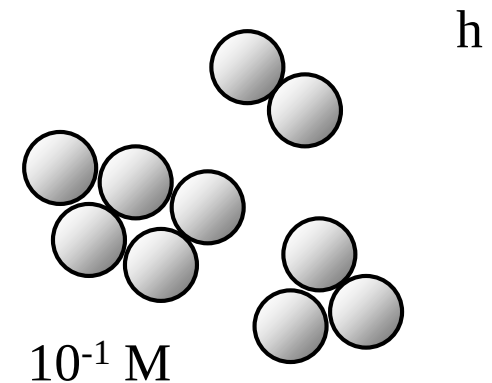
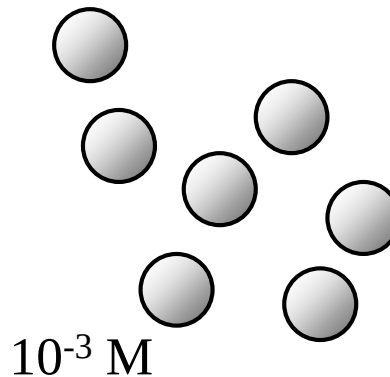
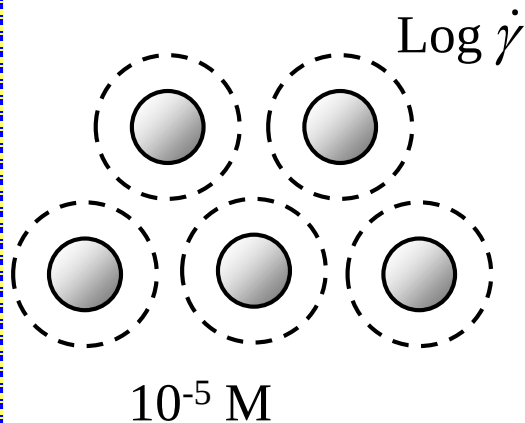
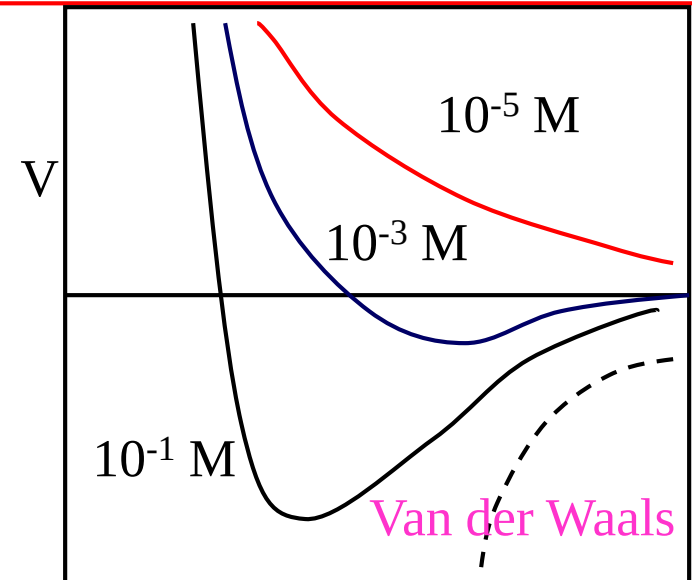
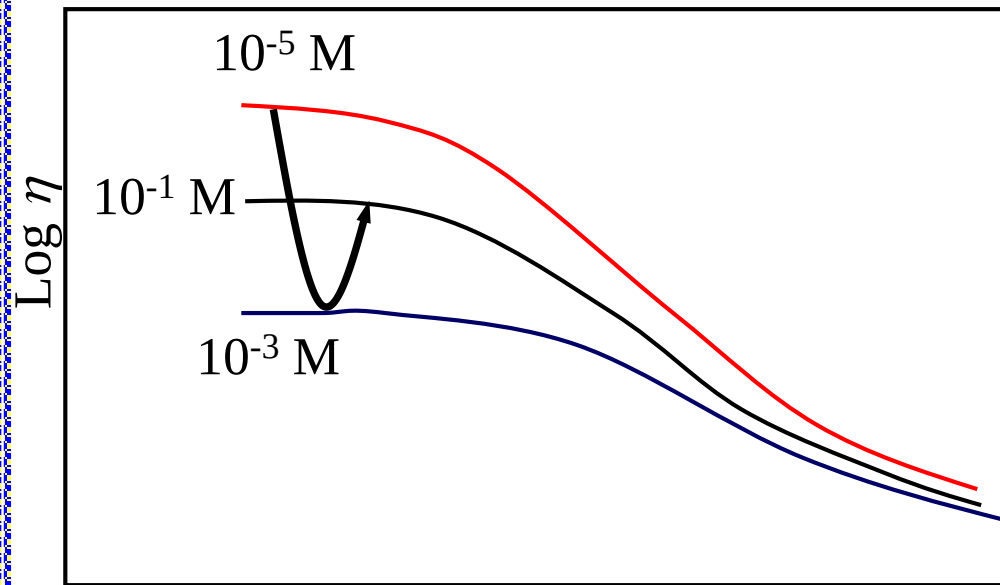
Lecture 22

ChE 467/596-012 Polymer Rheology

Overview

1. Review: particles with strong electrostatic repulsion
2. Sterically Stabilized Systems
3. Flocculated Systems
4. Research Examples

Particles with strong electrostatic repulsion: Electrolyte Effects



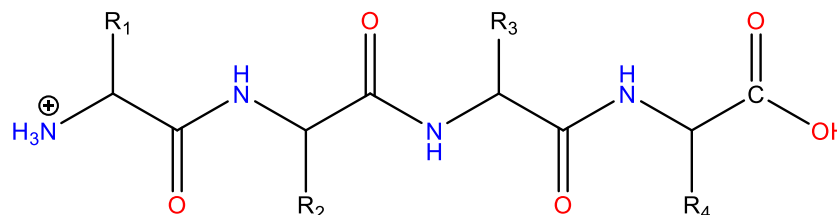
Increasing ionic strength



pH and Isoelectric Point

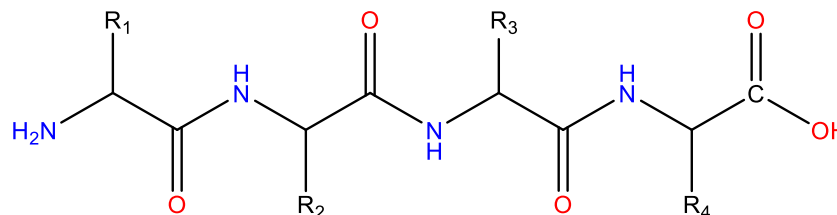
- $\text{pH} = -\log_{10}([\text{H}^+]) = \log_{10}\left(\frac{1}{[\text{H}^+]}\right)$
- What is the isoelectric point (IEP, pI)?
 - pH unit at which net charge on molecule = 0
 - Property of proteins based on amino acid sequence

1. $\text{pH} < \text{pI}$:



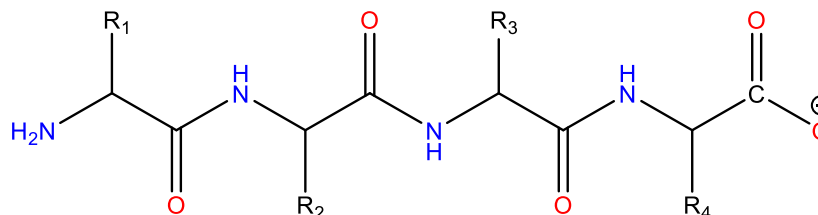
Net charge = +1

2. $\text{pH} = \text{pI}$:



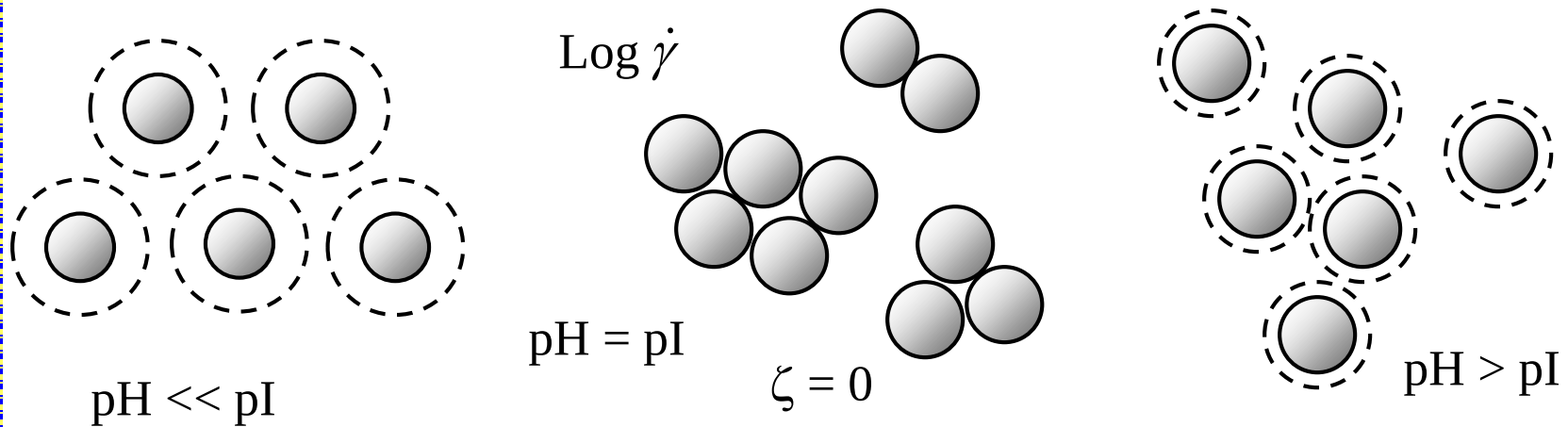
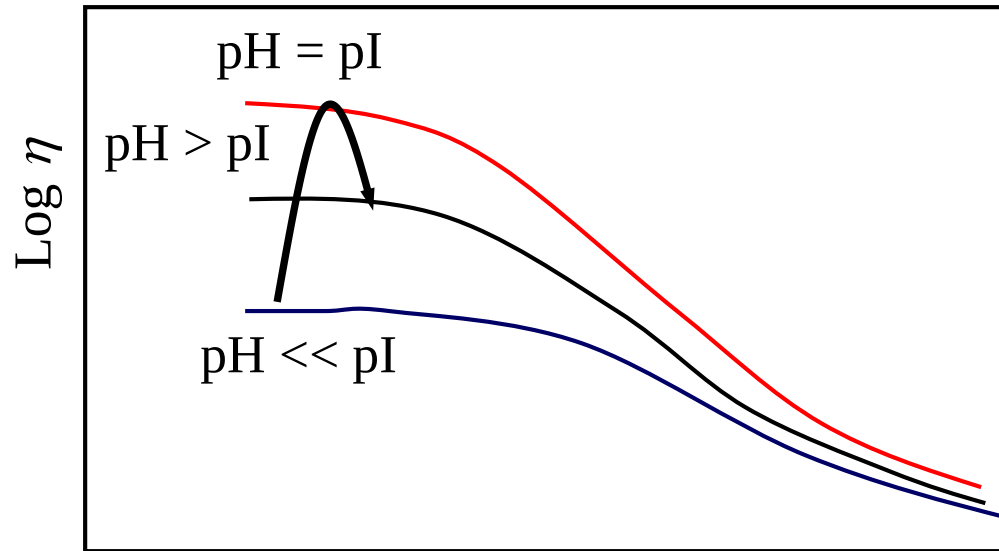
Net charge = 0

3. $\text{pH} > \text{pI}$:



Net charge = -1

Particles with strong electrostatic repulsion: pH Effects



Increasing pH →

In-Class Exercise: Effect of pH and Ionic Strength

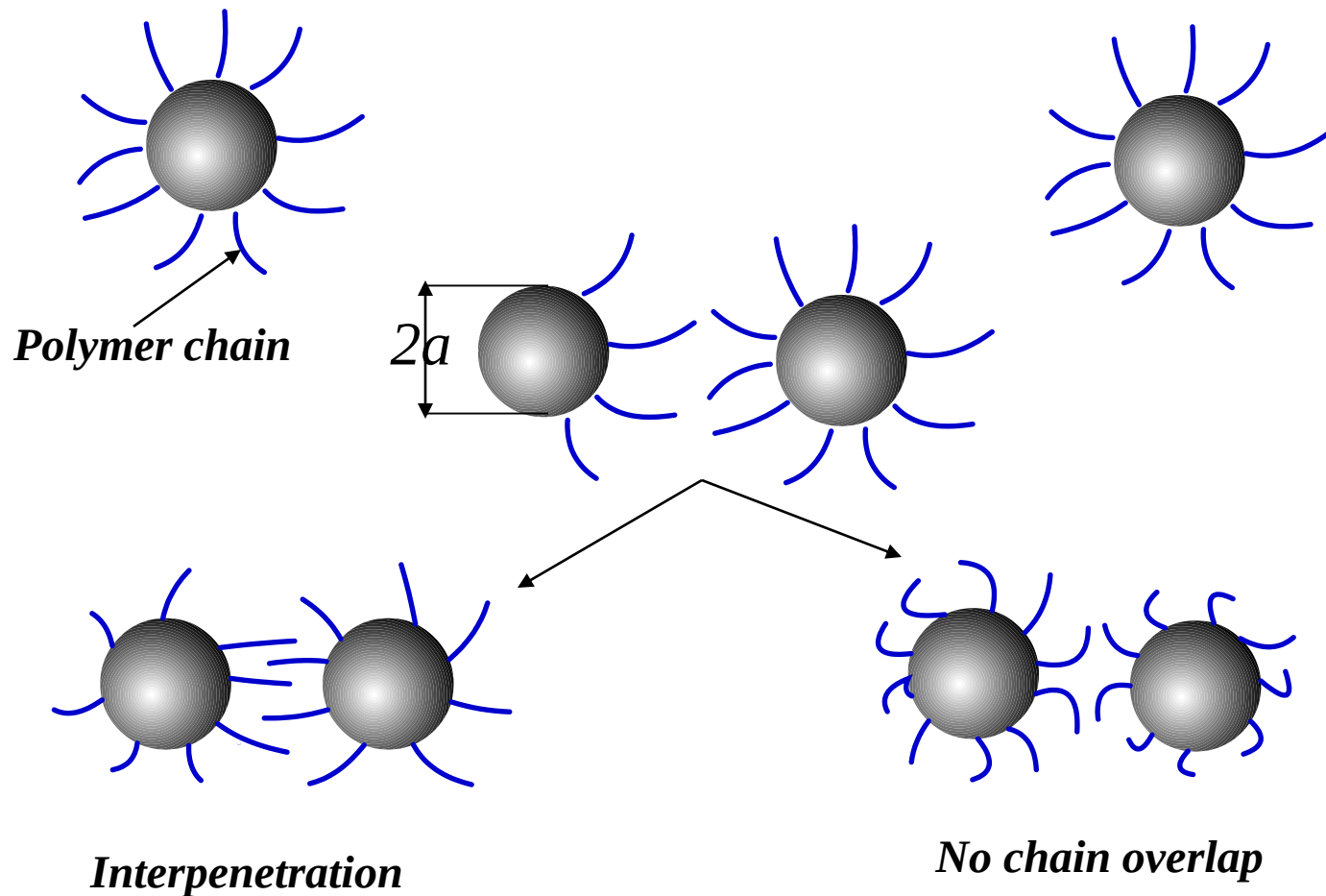
You have four solutions of 2 μm particles densely coated with a protein whose $\text{pI} = 5.2$:

- A. Solution $\text{pH} = 8.0$, ionic strength = 0.1 M
- B. Solution $\text{pH} = 8.0$, ionic strength = 0.01 M
- C. Solution $\text{pH} = 5.2$, ionic strength = 0.1 M
- D. Solution $\text{pH} = 5.2$, ionic strength = 0.01 M

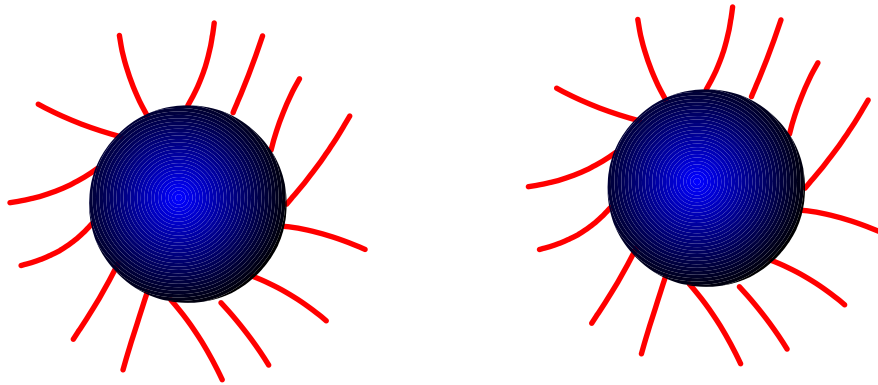
Rank the solutions in order of increasing η_o :

Answer:
 $B < A < C, D$

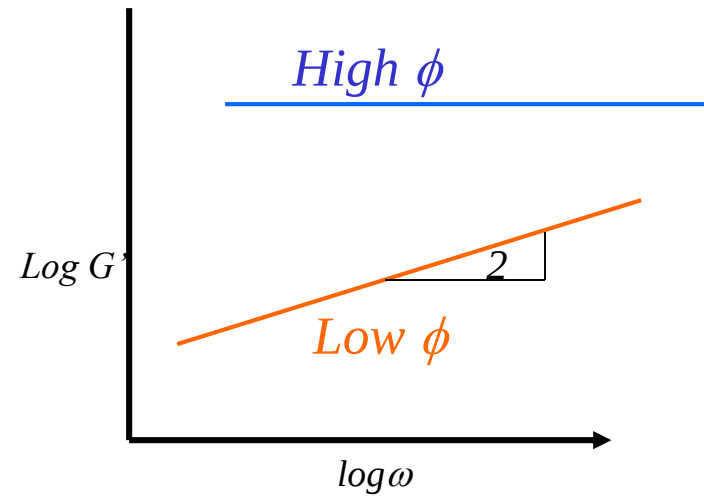
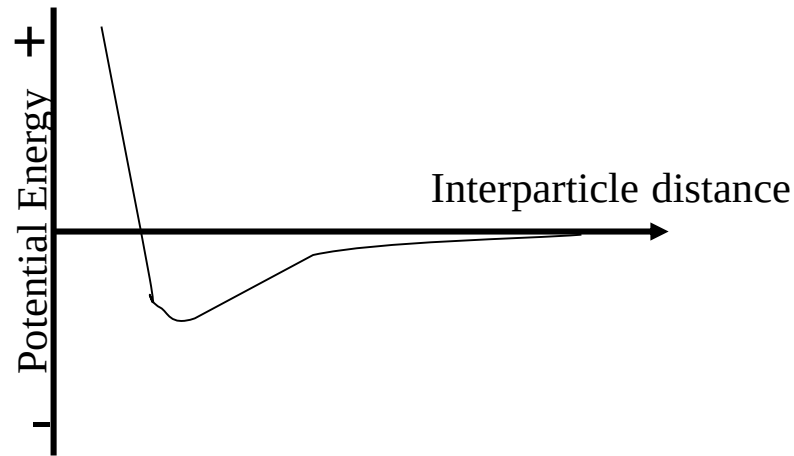
Sterically Stabilized System



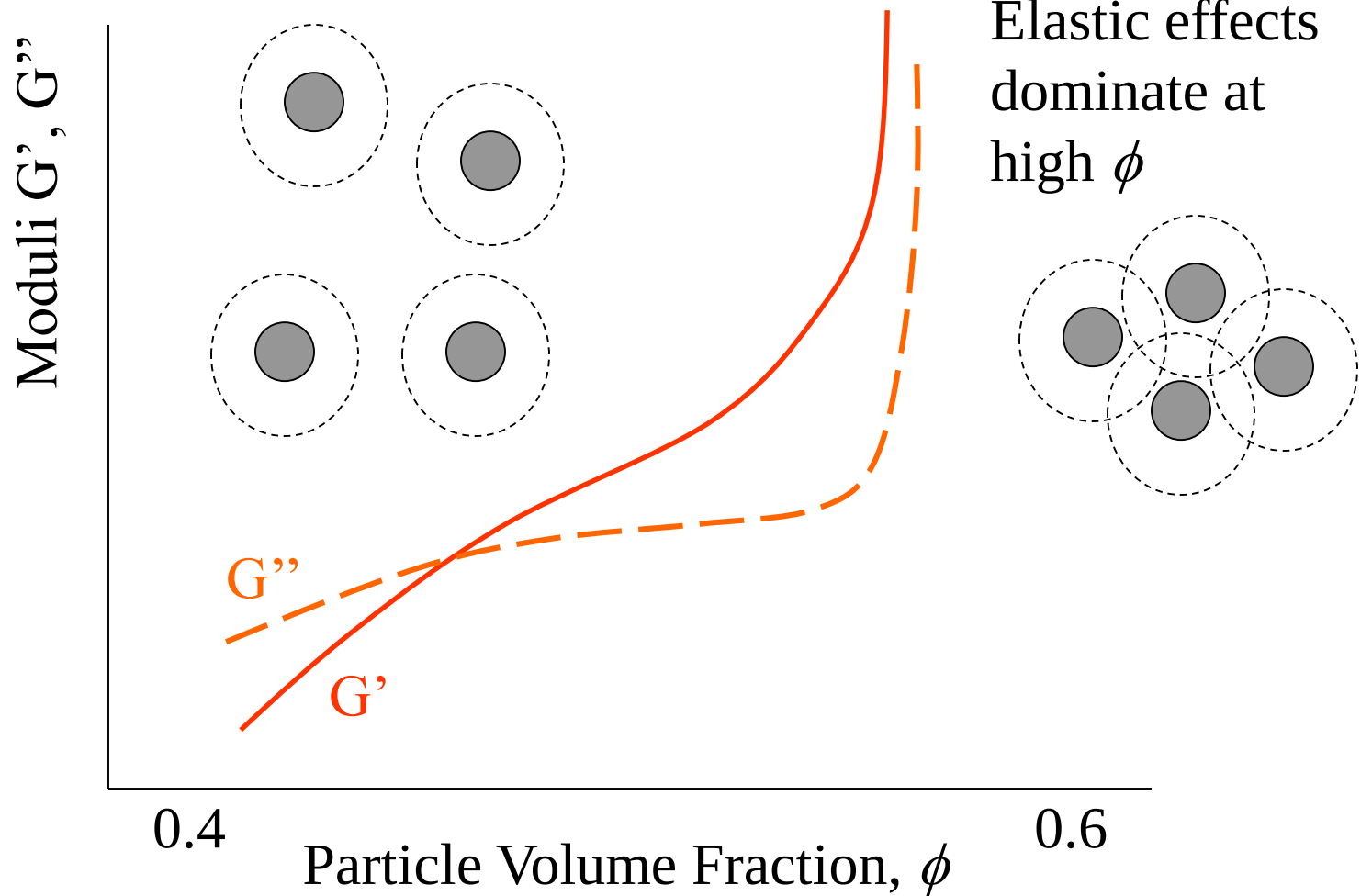
Polymeric or Sterically Stabilized System



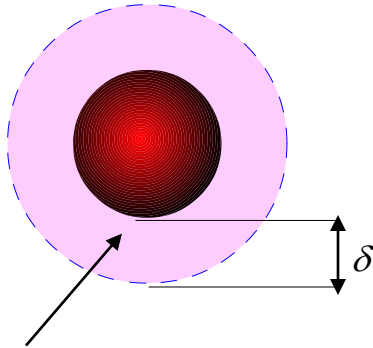
What is G' vs ω at different ϕ ?



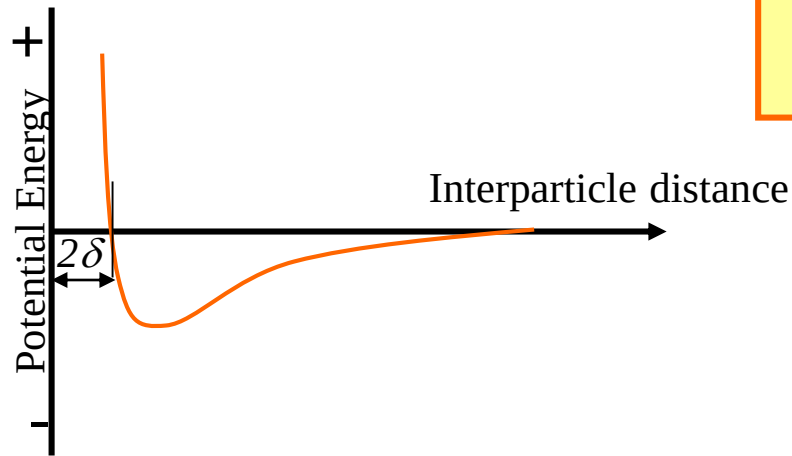
Dynamic Rheology - “Soft” Particles



Polymeric Stabilized System, cont'd

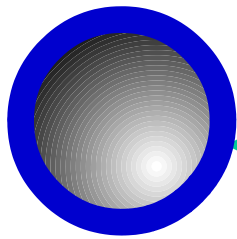


Layer full of polymer chains



- *Cannot penetrate*
- *Increased effective volume fraction of solid*
- *Starts to show Brownian Hard Sphere behavior*

Polymeric Stabilized System, cont'd



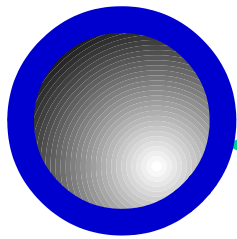
Layer thickness δ

- *Repulsions are “short” range*
- *Effectively increasing solid fraction*

$$\phi_{eff} = \phi \left(1 + \frac{\delta}{a} \right)^3$$

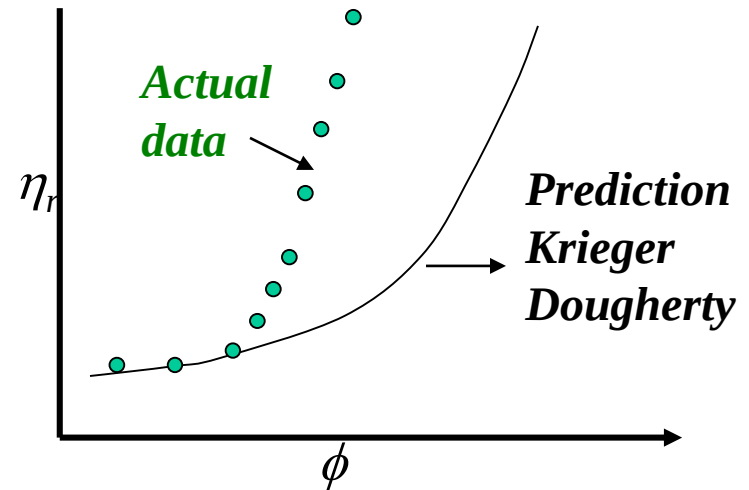
$$\eta_{rx}^{relative} = \left[1 - \frac{\phi_{eff}}{\phi_{max}} \right]^{-2.5\phi_{max}}$$

Polymeric Stabilized System, cont'd



Layer thickness δ

Back out δ from rheology data

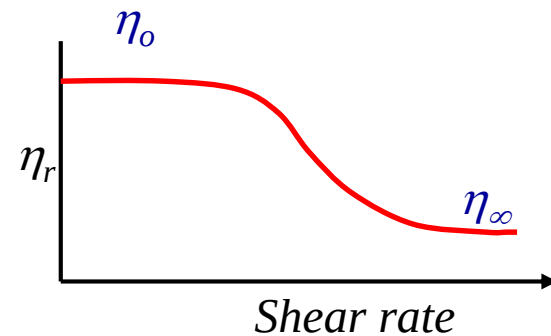
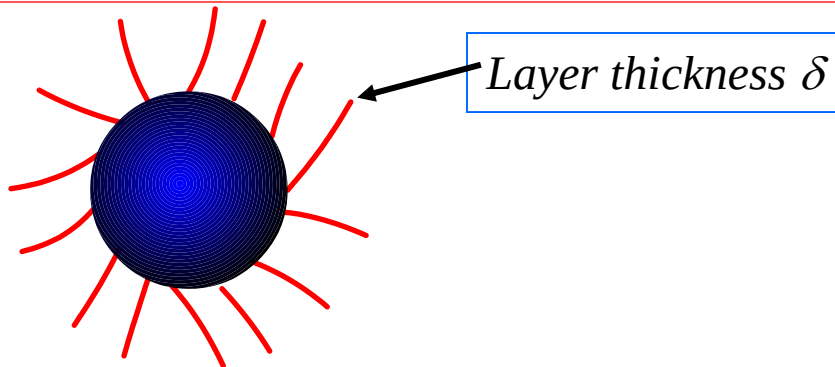


To get δ :

Use ϕ and adjust δ to match the experimental data with Krieger Dougherty eqn.

$$\phi_{\text{eff}} = \phi \left(1 + \frac{\delta}{a} \right)^3$$

In-class exercise



Given:

$$\eta_{ro} = 10$$

$$\Phi = 0.4$$

$$a = 100 \text{ nm}$$

Outline how would you find δ ?

$$\eta_{rx}^{\text{relative}} = \left[1 - \frac{\phi_{\text{eff}}}{\phi_{\text{max}}} \right]^{-2.5\phi_{\text{max}}}$$

Plug in for η and ϕ_{max} and get ϕ_{eff}

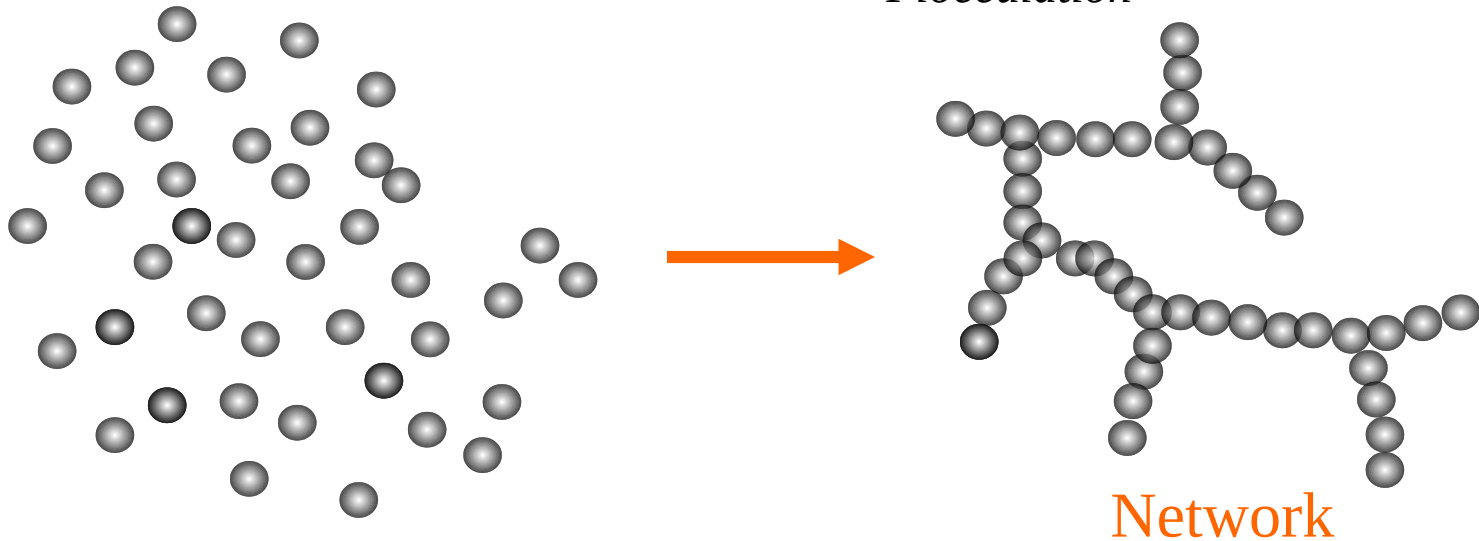
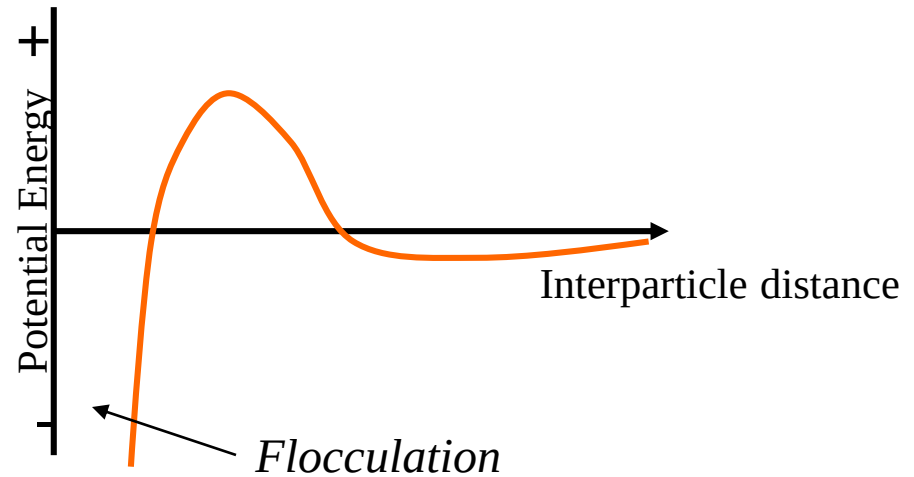
$$\phi_{\text{eff}} = \phi \left(1 + \frac{\delta}{a} \right)^3$$

Answer:

$$\Phi_{\text{eff}} = 0.49, \delta = 6.9 \text{ nm}$$

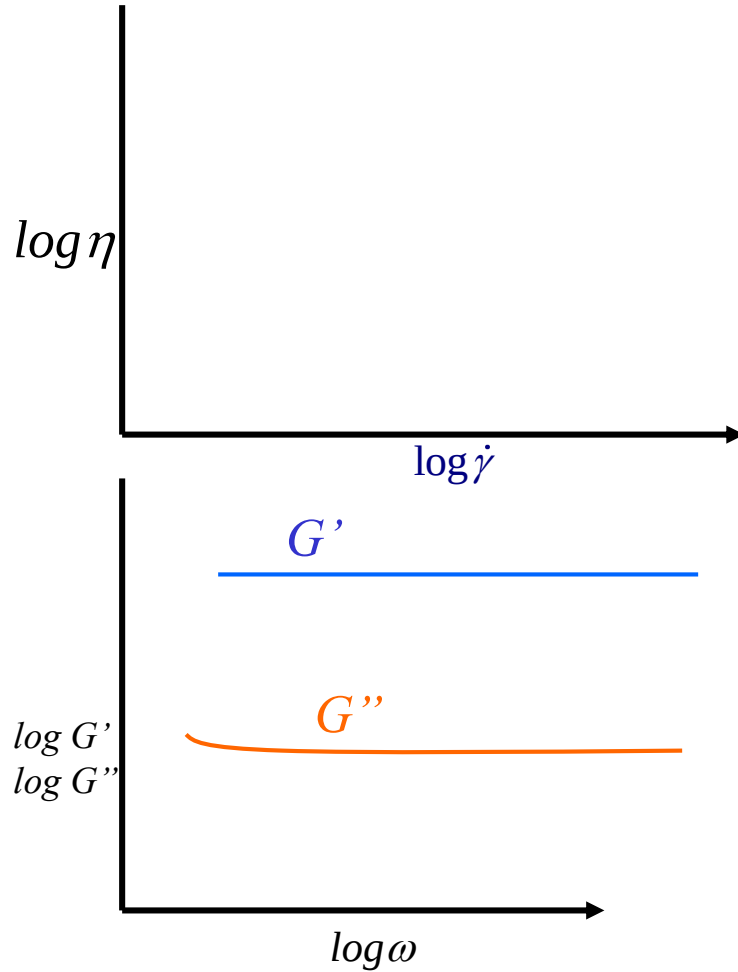
Flocculated Systems

- *Thixotropy*
- *Physical gels*

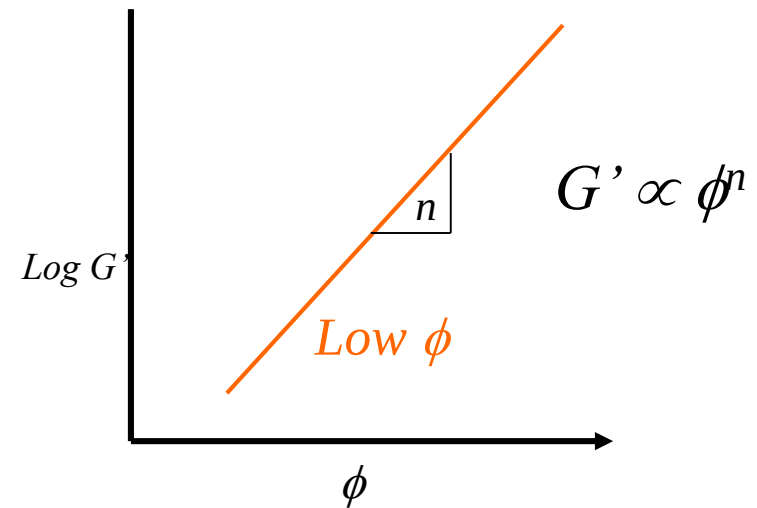
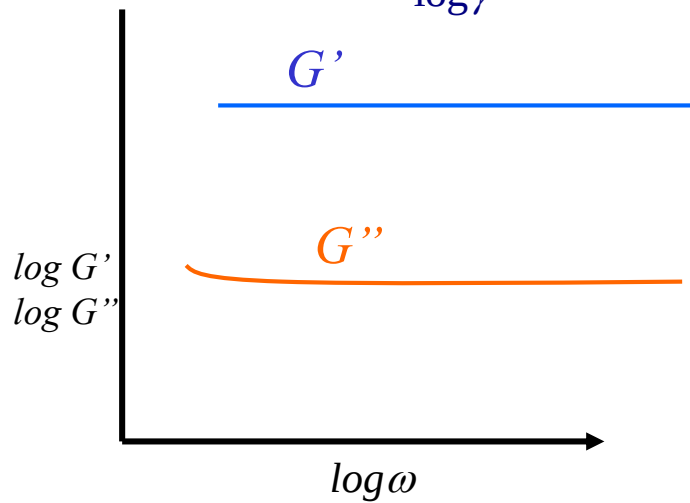
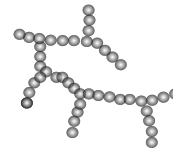
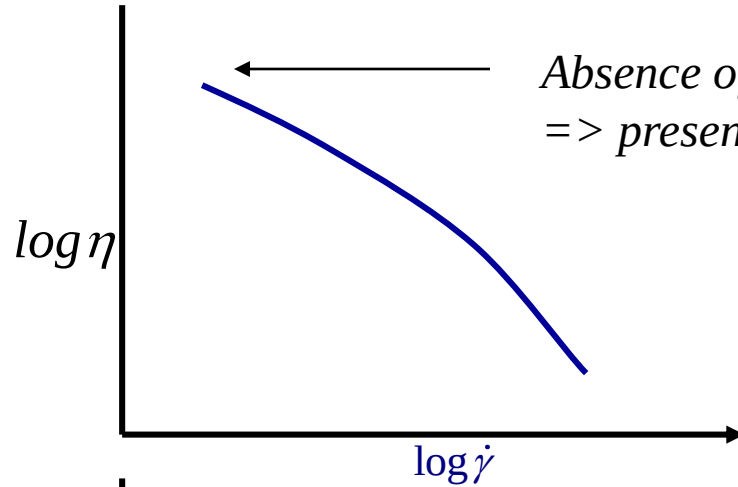


Flocculated Systems, cont'd

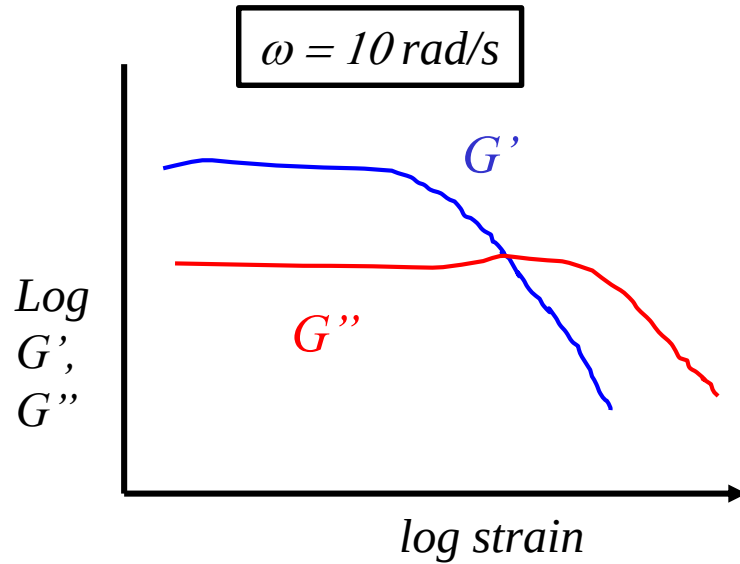
Plot the viscosity versus shear rate behavior of this sample



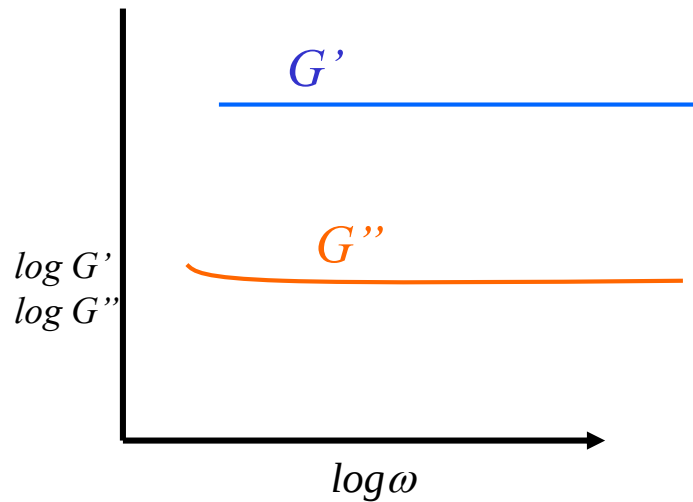
Flocculated Systems, cont'd



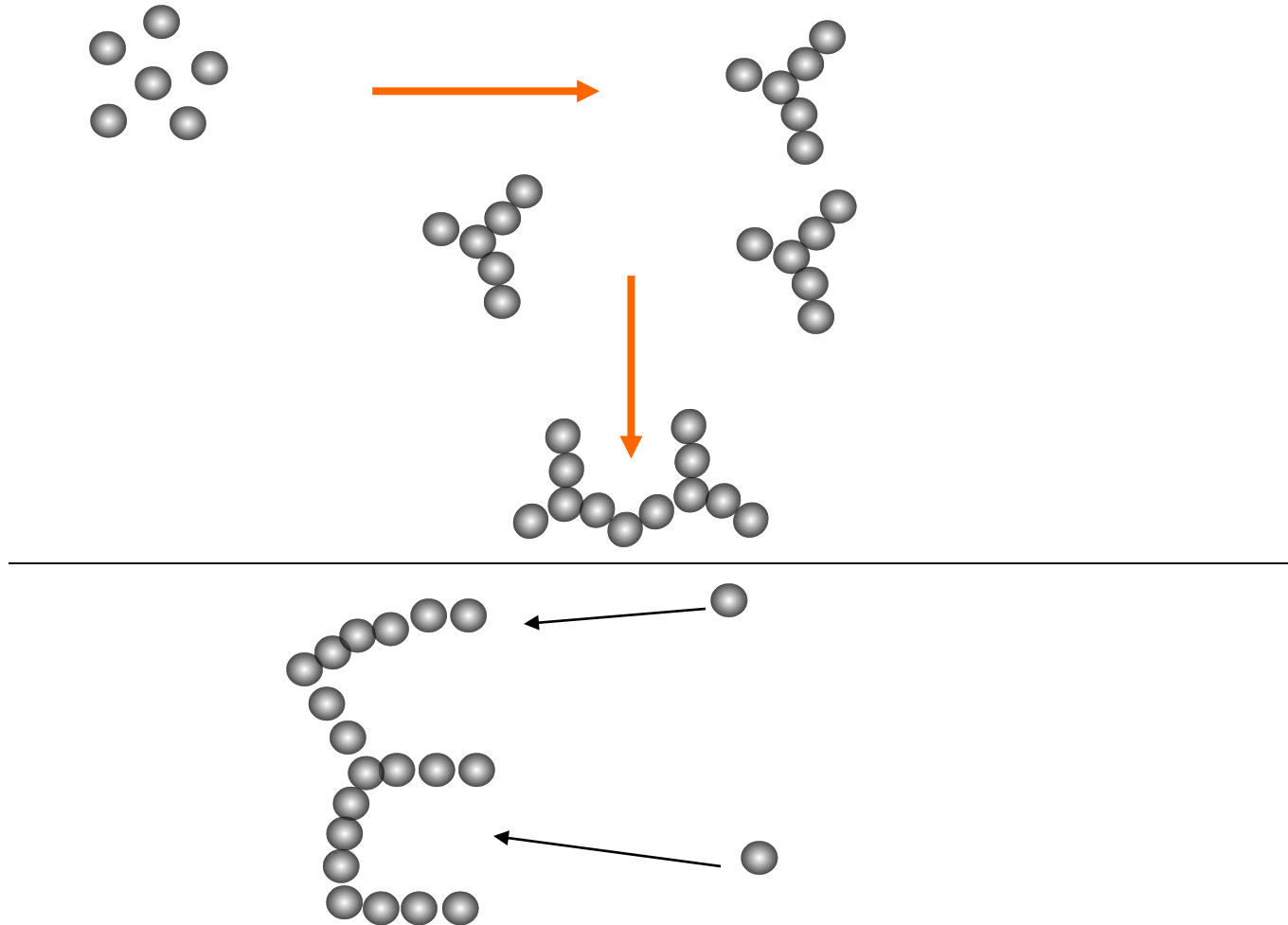
Flocculated Systems, cont'd



Plot the modulus versus strain behavior



Flocculated Systems, cont'd

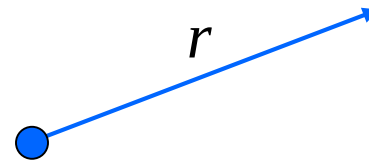


Flocculated Systems, cont'd

Fractal nature in many of these systems

$$m \text{ (mass) or } n \text{ (number)} \sim r^{d_f}$$

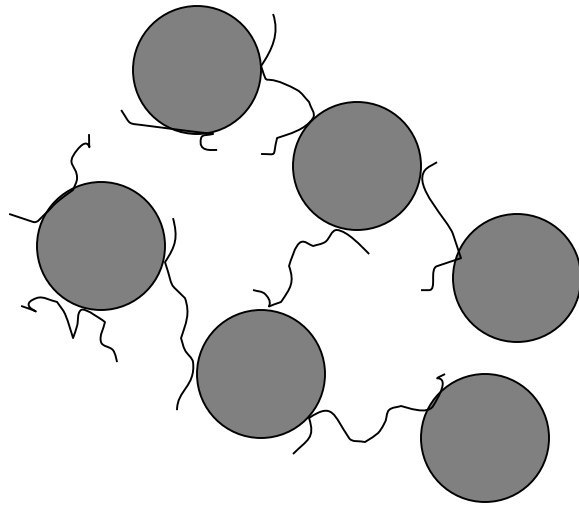
$d_f = \text{Fractal dimension} \leq 3$



$d_f \approx 1.8$, aggregation mechanism = **Diffusion-limited**
(incoming particle sticks to periphery of floc,
high “sticking probability”)

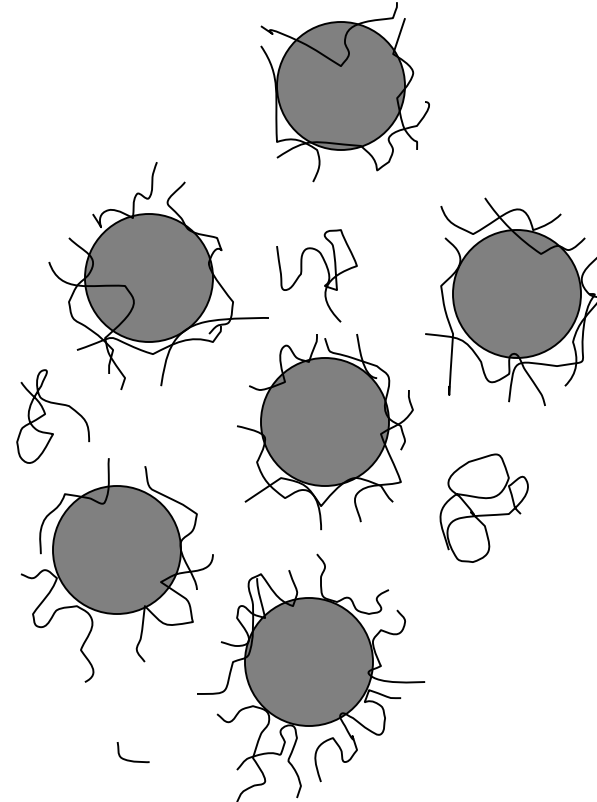
$d_f \approx 2.1\text{-}2.2$, aggregation mechanism = **Reaction-limited**
(incoming particle bounces around before finding a home,
low “sticking probability”)

Flocculation from Adsorbing Polymer



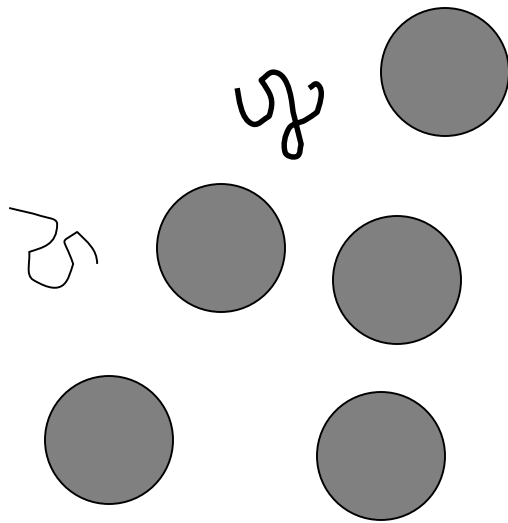
Low polymer concentration
Flocculation by bridging
mechanism

Low surface coverage



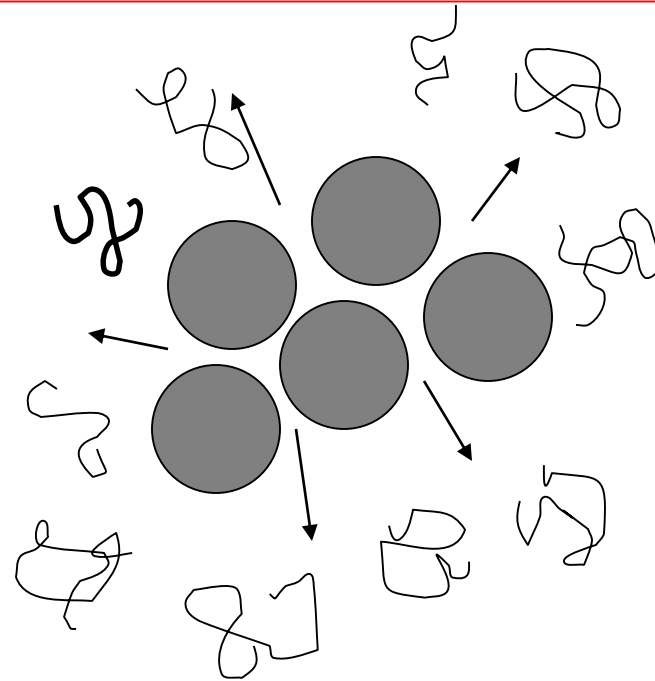
High polymer concentration
Stabilization

Non-adsorbing polymer - Depletion Flocculation



Low polymer concentration

Stable Dispersion



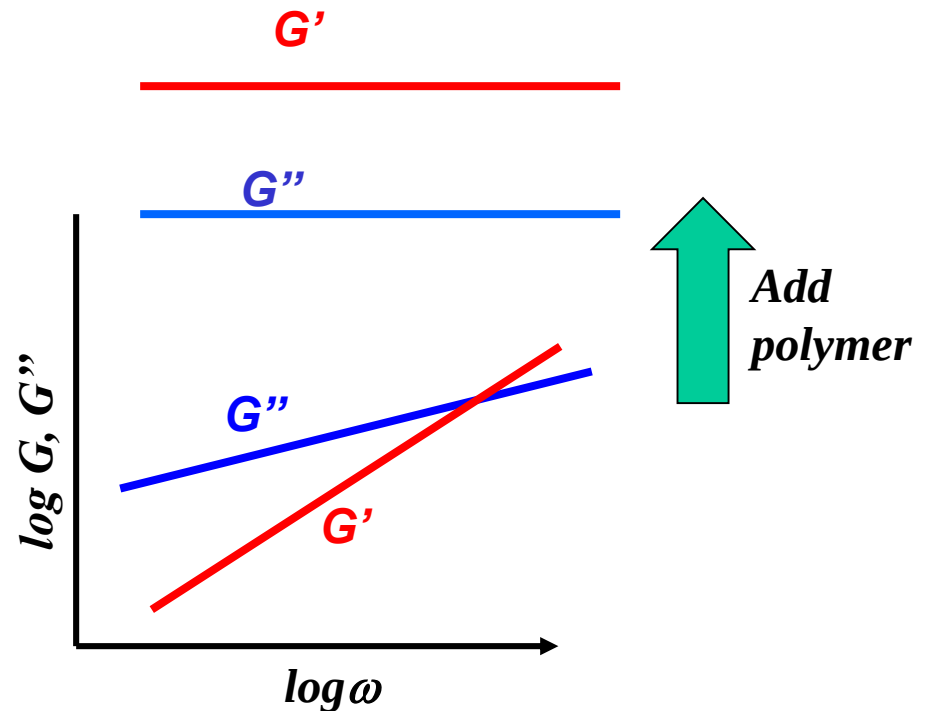
High polymer concentration
Flocculation by volume
restriction mechanism

$$\phi > \phi_p^+$$

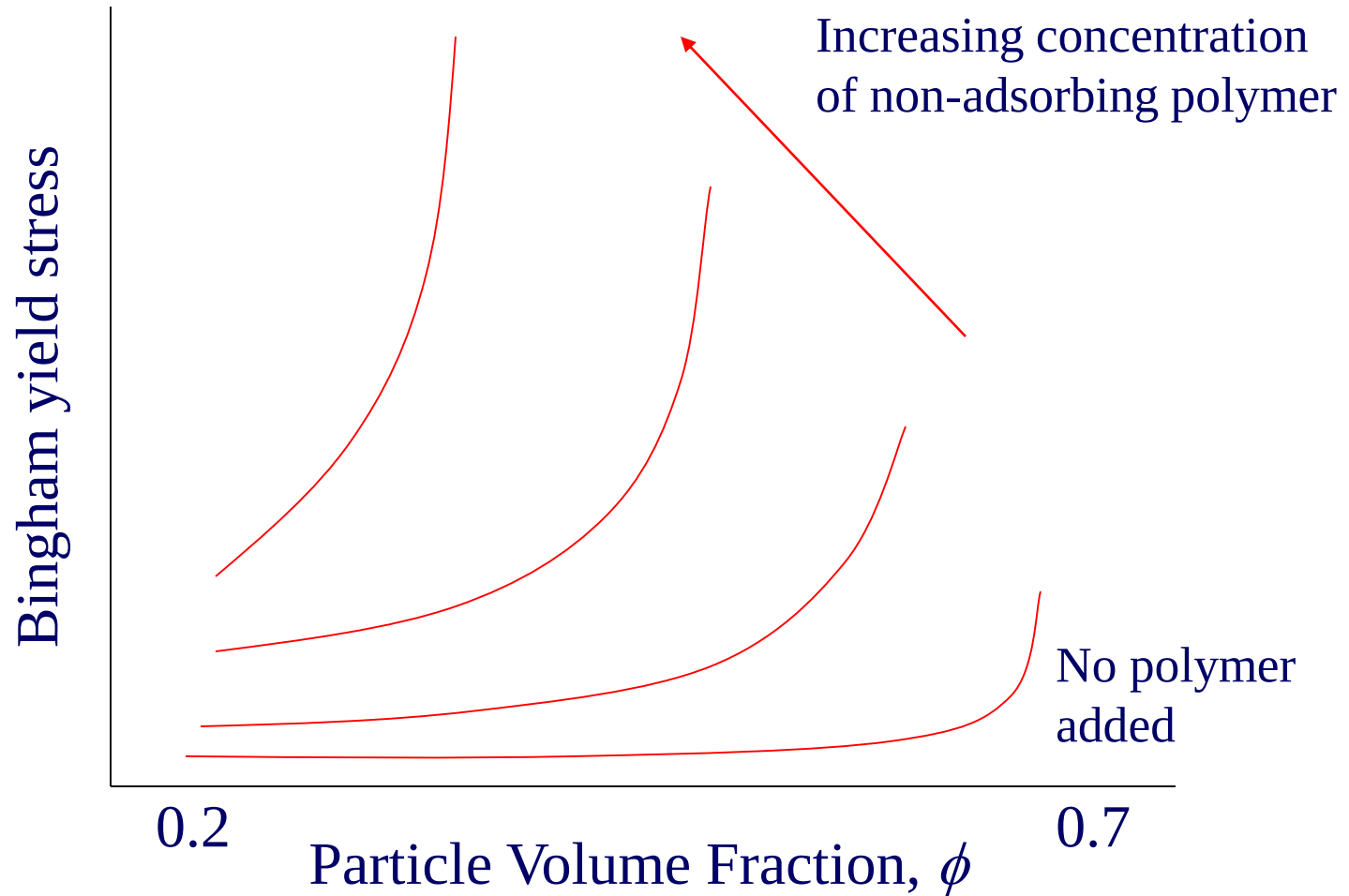
Depletion Flocculation and gelation

How can you determine that gelation is due to depletion flocculation?

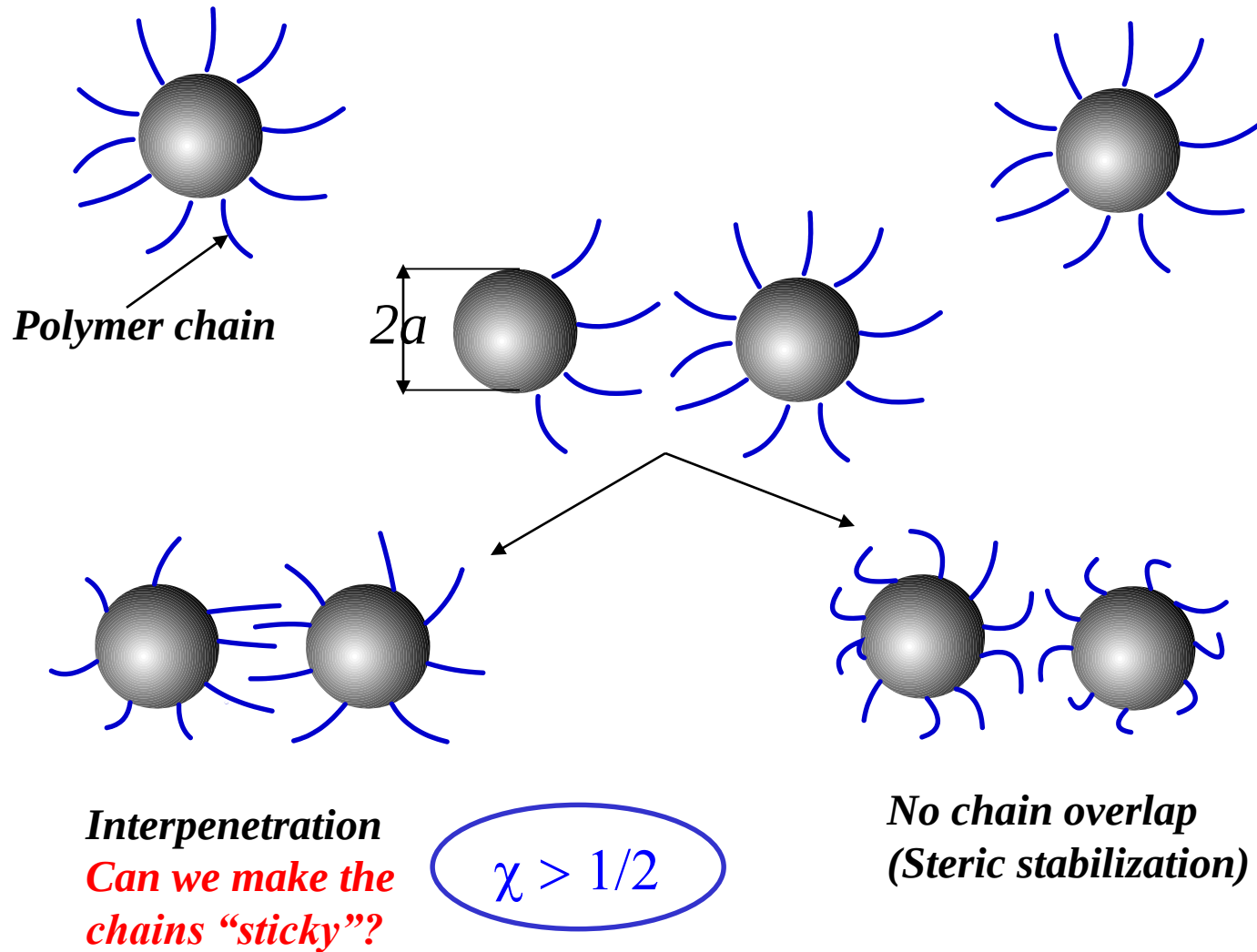
Add polymer to suspension and see if gelation occurs



Depletion Flocculation and gelation

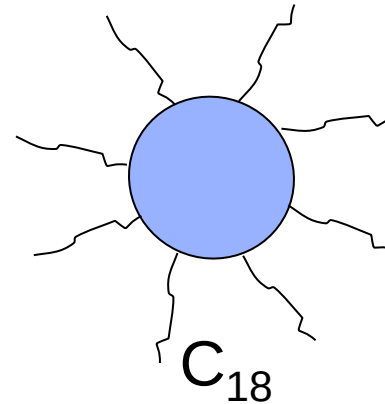
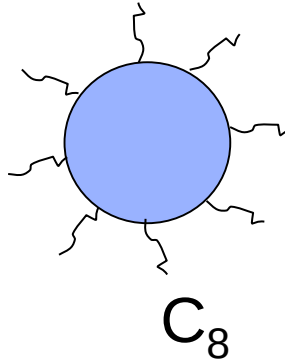
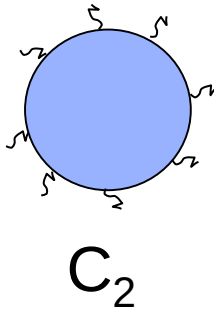


Polymer Grafted System

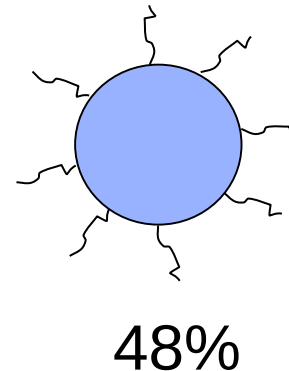
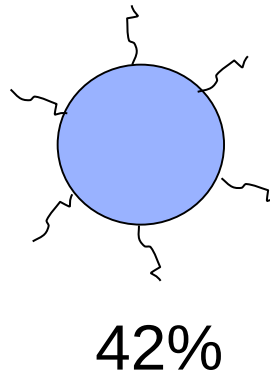
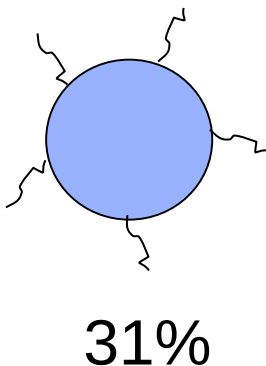


Range of Fumed Silicas with Non-Polar Surfaces

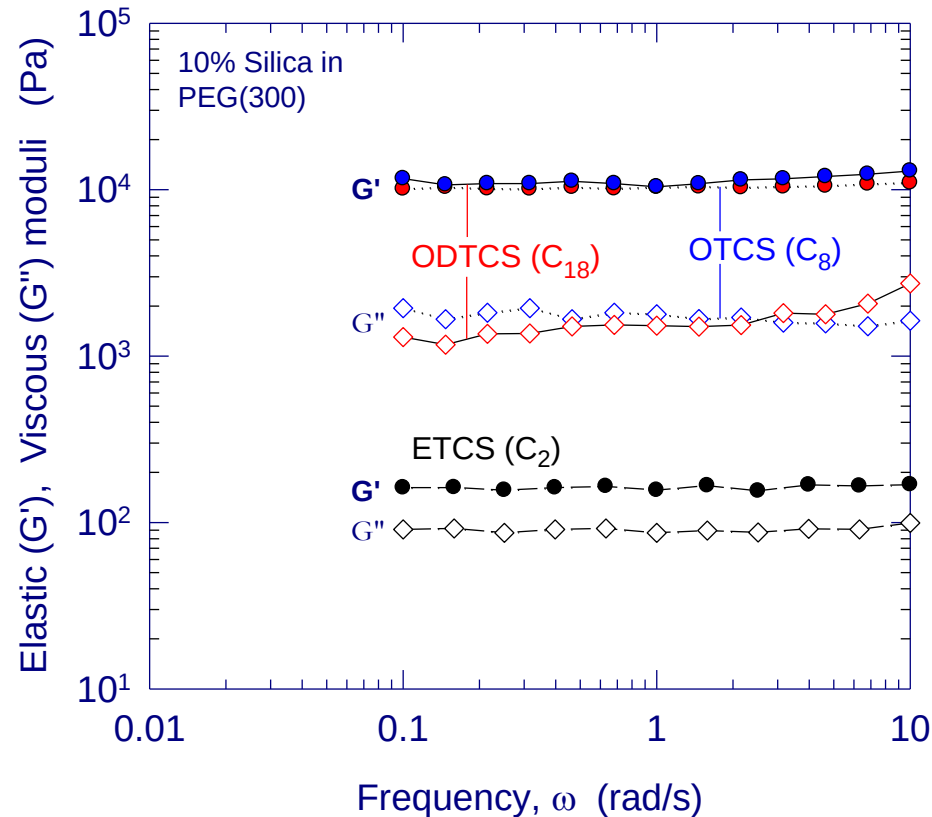
Different Chain Lengths



Different Surface Coverage by Alkyl Chains (C_8)

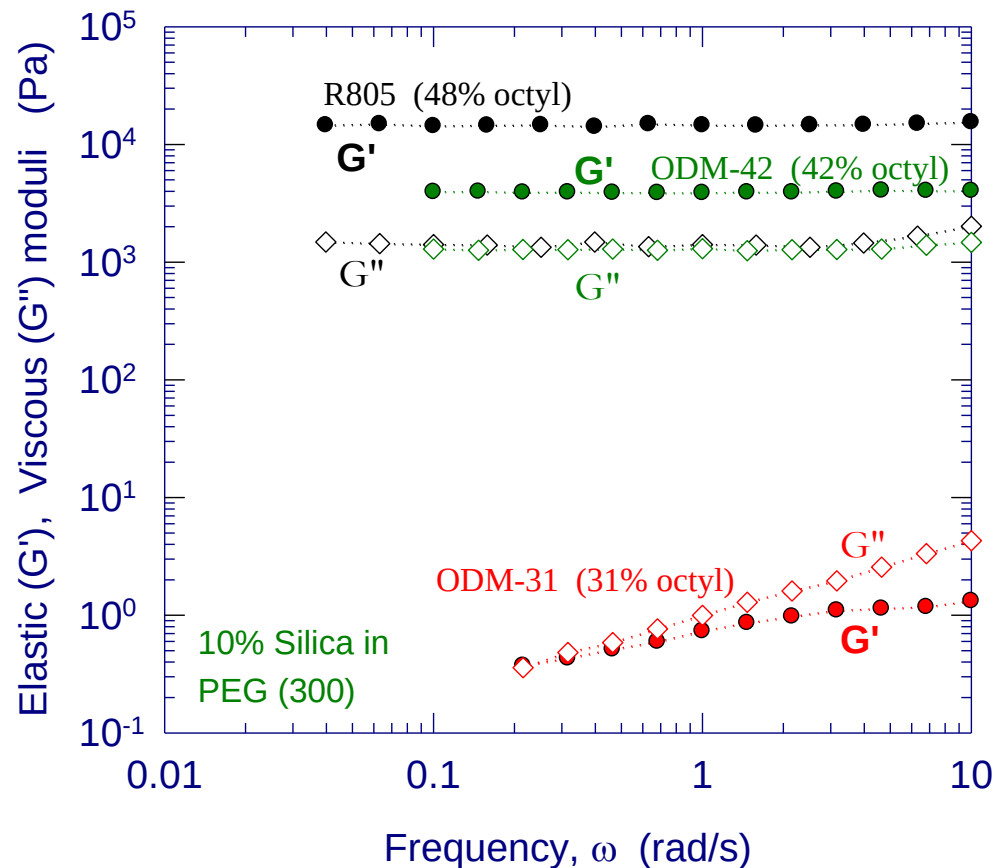


Varying Chain Length of *n*-alkyl Groups on FS



- Elastic network in all cases
- Degree of gel formation **increases** with chain length
- No enhancement in modulus beyond certain chain length (C_8)

Surface Coverage Effect on octyl-modified FS



- Flocculated suspension at lowest surface coverage
- Gel-like behavior **at high surface coverage**
- Shows that high coverage of octyl chains shields silanols

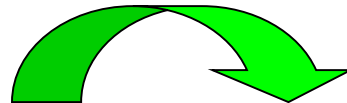
Interactions Between Tethered chains

Interactions between tethered chains depend on:

segment – segment contacts
+
solvent – solvent contacts

vs.

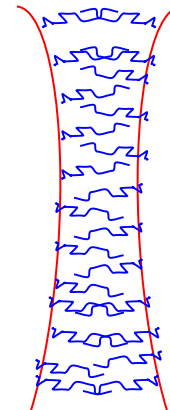
segment - solvent
contacts



Governed by χ parameter

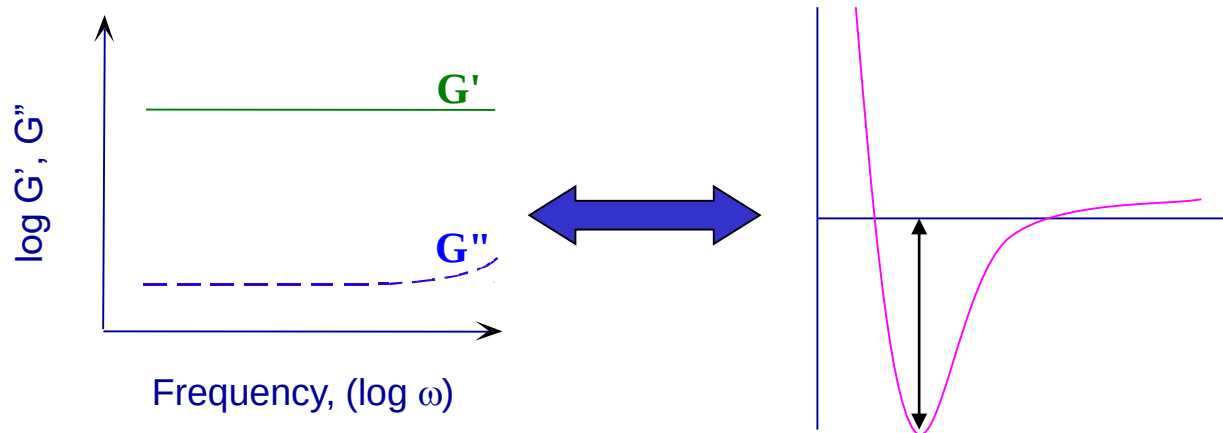
Mismatch between chains & solvent $\Rightarrow \chi > 1/2$

- Chains on particles become “sticky” (attractive) at close approach



Colloidal Interactions and Rheology

G' related to depth of attractive potential well



We obtain the following scaling relationship: (Russel, 1989) $G' \sim N \cdot V''_{\min}$

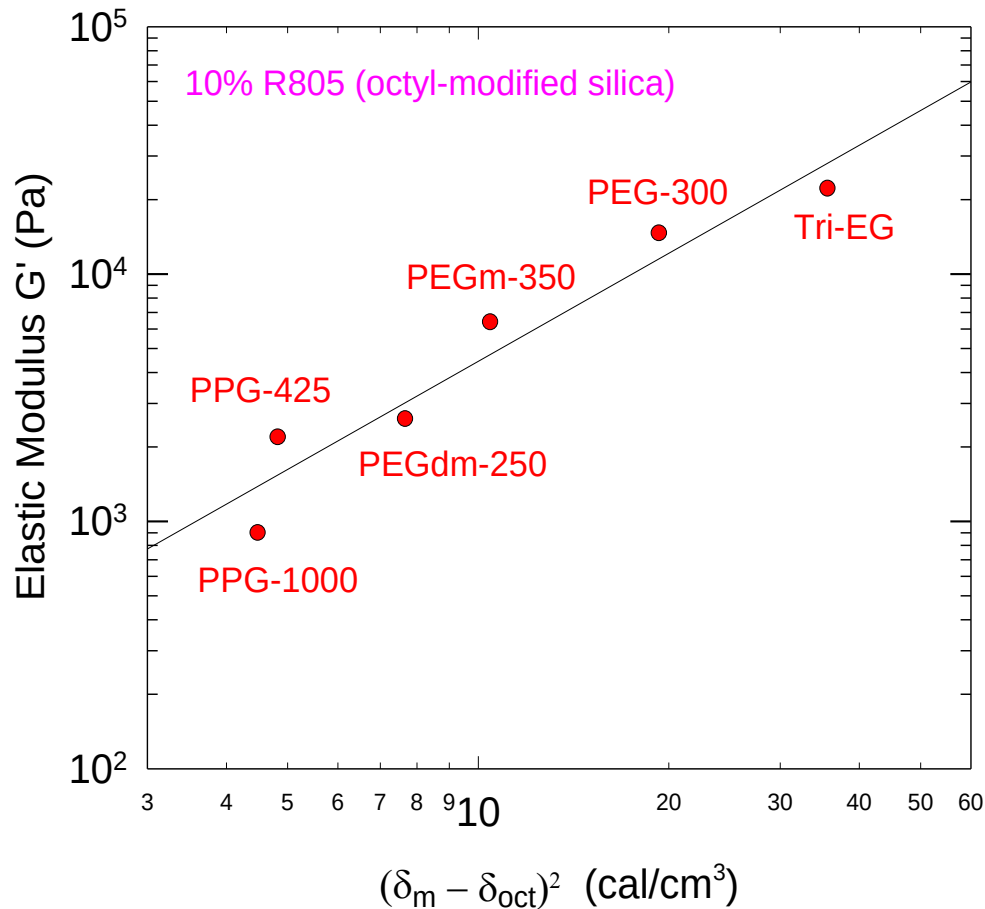
Since $\frac{V}{kT} \sim (\frac{1}{2} - \chi)$

and χ can be related to solubility parameters: $\chi = \frac{V_m}{RT} (\delta_s - \delta_m)^2 + 0.34$

we finally get:

$$G' \sim (\delta_s - \delta_m)^2$$

Colloidal Interactions of Fumed Silica Systems



$$G' \sim (\delta_s - \delta_m)^2$$

Incompatibility between surface chains and liquid molecules induces chains on adjacent particles to interpenetrate

Elastic modulus scales with the difference in solubility parameter between the liquid medium (δ_m) and silica surface (δ_s)

Raghavan, et al, *Langmuir* (2000)

30 of 30

- Review of Multiphase Systems
 - How is viscosity affected in foams and emulsions?
 - What factors are important for dilute suspensions?
 - How does a non-dilute suspension differ?
- In class problems